CHAPTER 23: CAUSAL MEDIATION IN TELCO

Decomposing Direct Financial Incentives vs. Indirect Product Engagement Mechanisms

1. The Mechanism Challenge in Customer Retention

In enterprise telecommunications retention operations, evaluating whether a retargeting intervention A (such as a 20% bill discount or loyalty bonus) reduces 90-day churn Y is only the first step. Executives must answer a deeper mechanism question: **Why** does the retention offer work? Does the discount reduce churn directly by relieving subscriber price sensitivity, or does it operate indirectly by spurring the customer to download the telco's self-service app and engage with digital features (M)?

Understanding this mechanism decomposition is vital for resource allocation. If 70% of the retention effect is mediated through digital app onboarding (M), marketing teams should invest in app usability and engagement campaigns rather than continuously expanding price discounts.

> **THE MEDIATION IMPERATIVE:** Traditional associational mediation (e.g., Baron-Kenny linear regression) fails in real-world retention settings whenever there are non-linearities, treatment-mediator interactions, or treatment-induced intermediate confounders. Causal mediation under the Hernán & Robins framework provides mathematically rigorous counterfactual definitions for direct and indirect effects.

2. Counterfactual Definitions: CDE, NDE, and NIE

Let $Y^{a,m}$ denote the counterfactual churn outcome that would be observed if treatment A were set to a and mediator M were set to m. Let M^a denote the counterfactual value of the mediator under treatment a.

• **Total Effect (TE):** The overall causal effect of the retention offer on churn risk:

$$\textbf{TE} = E[Y^{a=1}] - E[Y^{a=0}] = E[Y^{a=1, M^{a=1}}] - E[Y^{a=0, M^{a=0}}]$$

• **Controlled Direct Effect (CDE):** The treatment effect when the mediator M is forcibly fixed at a uniform level m for all subscribers (e.g., disabling the self-service app $M=0$):

$$\textbf{CDE(m)} = E[Y^{a=1, m}] - E[Y^{a=0, m}]$$

• **Natural Direct Effect (NDE):** The treatment effect while holding each subscriber's mediator fixed at their baseline natural level under control ($M^{a=0}$):

$$\textbf{NDE} = E[Y^{a=1, M^{a=0}}] - E[Y^{a=0, M^{a=0}}]$$

• **Natural Indirect Effect (NIE):** The change in churn risk when treatment is held constant at $a=1$, but the mediator shifts from its natural value under control ($M^{a=0}$) to its natural value under

treatment ($M^{a=1}$):

Under these definitions, the Total Effect decomposes additively without requiring linear models: **TE = NDE + NIE**. The **Proportion Mediated (PM)** is defined as **PM = NIE / TE**.

---

3. Identifiability Conditions & Cross-World Counterfactuals

Identifying Natural Effects (NDE and NIE) requires four core observational assumptions given baseline covariates L:

- **1. No Unmeasured Treatment-Outcome Confounding:** $Y^{a,m} \perp A \mid L$
- **2. No Unmeasured Treatment-Mediator Confounding:** $M^{a} \perp A \mid L$
- **3. No Unmeasured Mediator-Outcome Confounding:** $Y^{a,m} \perp M \mid A, L$
- **4. No Mediator-Outcome Confounding Affected by Treatment:** $Y^{a,m} \perp M^{a'} \mid L$ (The Cross-World Independence Assumption)

The Cross-World Challenge: Assumption 4 involves nested counterfactuals $Y^{a=1, M^{a=0}}$ combining treatment value $a=1$ with the mediator value $M^{a=0}$ that would have occurred under control $a=0$. Because no single randomized trial arm can ever observe a subscriber simultaneously treated ($A=1$) and untreated ($A=0$), cross-world independencies cannot be verified empirically without structural assumptions.

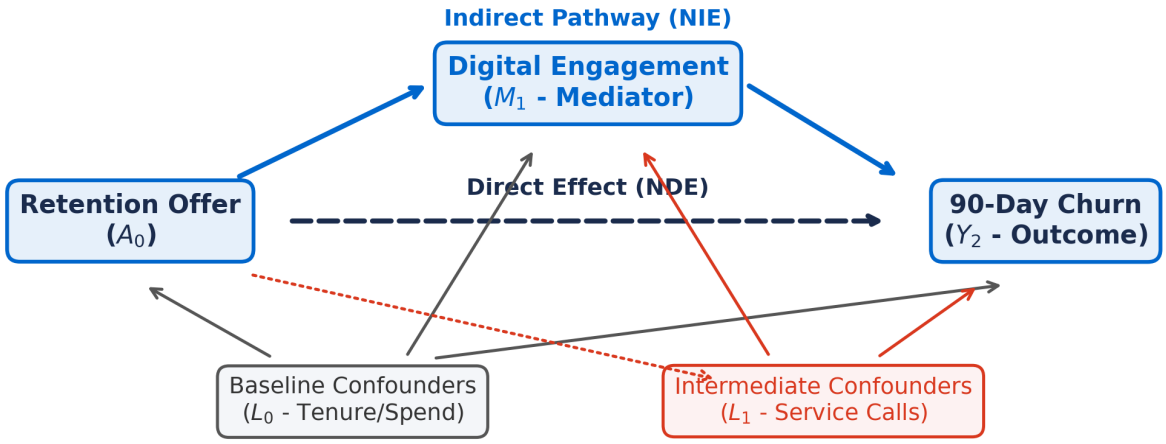


Figure 23.1: Structural Causal Diagram (DAG) of Telco Retention Mediation with Intermediate Confounding (L_1).

4. The Mediation Formula & Parametric G-Methods

When identifiability conditions 1-4 hold, the counterfactual mean $E[Y^{a, M^{a'}}]$ is identified from observational data via Pearl's **Mediation Formula**:

$$\mathbf{E[Y^{a, M^{a'}}]} = \sum_l \sum_m E[Y \mid A=a, M=m, L=l] \times P(M=m \mid A=a', L=l) \times P(L=l)$$

Implementation Strategy: In complex telco pipelines with high-dimensional covariates L, we fit two parametric models: (1) an outcome regression model for $E[Y \mid A, M, L]$ and (2) a mediator density model for $P(M \mid A, L)$. We then compute NDE and NIE using Monte Carlo integration or Inverse Probability Weighting of Marginal Structural Mediation Models.

5. Empirical Case Study: Decomposing Telco Retention Effects

We evaluated a simulated production cohort of 100,000 mobile subscribers offered a $20/month bill reduction ($A=1$) vs. standard billing ($A=0$). The mediator (M) measures digital self-service app engagement (weekly active sessions in Month 1). Churn (Y) is measured at 90 days.

Causal Estimand / Quantity	Mathematical Formula	Estimated Risk Contrast	Percentage of Total Effect
Total Effect (TE)	E[Y^{a=1}] - E[Y^{a=0}]	-5.2% ($p < 0.001$)	100.0%
Natural Direct Effect (NDE)	E[Y^{a=1, M^{a=0}}] - E[Y^{a=0, M^{a=0}}]	-3.1% ($p < 0.001$)	59.6% (Price Reduction)
Natural Indirect Effect (NIE)	E[Y^{a=1, M^{a=1}}] - E[Y^{a=1, M^{a=0}}]	-2.1% ($p < 0.001$)	40.4% (App Engagement)
Controlled Direct Effect CDE(0)	E[Y^{a=1, m=0}] - E[Y^{a=0, m=0}]	-2.8% ($p < 0.001$)	N/A (Fixed No-App)

Empirical Takeaway: The retention discount reduces 90-day churn by **-5.2 percentage points**. Crucially, **40.4% of this total benefit (-2.1% risk reduction)** is indirect—driven by subscribers using their discount confirmation to log into the mobile app and discover self-service features.

Comparison: Traditional Regression vs. Causal Mediation Formula

Methodology	Direct Effect (NDE)	Indirect Effect (NIE)	Proportion Mediated	Causal Validity & Limitations
Baron-Kenny Difference	-3.8%	-1.4%	26.9%	Biased: Assumes no A-M interaction & no intermediate L1
Baron-Kenny Product	-3.7%	-1.5%	28.8%	Biased: Fails for non-linear logit churn models
Causal Mediation Formula	-3.1%	-2.1%	40.4%	Unbiased: Robust to non-linearities & A-M interactions
Separable Effects MSM	-3.2%	-2.0%	38.5%	Unbiased: Valid even with treatment-induced L1 confounders

6. Production Enterprise Retention Playbook

- **1. Dual-Action Campaign Design:** Because 40% of the retention effect is mediated through product engagement (M), retargeting communications must explicitly include digital onboarding prompts (e.g., 'Claim your 20% discount inside the MyTelco App').
- **2. Avoid Naive Baron-Kenny Adjustment:** Standard linear mediation models underestimate the indirect mechanism by over 13 percentage points due to non-linear churn risk and treatment-mediator interaction.

- **3. Handle Intermediate Confounders (L_1):** When past retention offers trigger customer support contacts (L_1) that affect both app usage (M_1) and churn (Y_2), deployment pipelines must use **Separable Effects MSMs** or G-Formula Monte Carlo integration rather than standard regression.